Bella Boné
Portfolio Final Report
C1 Data Analytics Deliverable #7
2026

Introduction

In the year 2026 and beyond, artificial intelligence continues to change the nature of work. Understanding which occupations may be most exposed to AI can help workers, employers, and policymakers anticipate changes in the labor market. The following analysis examines AI Occupational Exposure (AIOE), wages, projected employment growth, and occupation categories to identify patterns associated with AIOE.

The analysis at hand examined how AIOE, projected employment growth, wages, occupational categories, and education requirements interact across U.S. occupations. Using BLS Employment Projections data alongside AIOE measures, the analysis identifies patterns that can help distinguish between occupations that are simply exposed to AI-driven change and those that may face greater economic vulnerability. The visualizations explore these relationships to better understand which factors may contribute to career resilience in an increasingly automated workforce.The research questions for the project were as follows :

1. What is the relationship between an occupation's average earning, projected employment growth, and AI exposure?
2. Do occupations with higher AI exposure share particular wage categories or occupational characteristics?
3. How can AI exposure , wages, and projected employment growth be used together to identify occupations that may be more or less vulnerable to automation and AI ?

Business Impact

The analysis at hand provides insights for businesses navigating the evolving impact of artificial intelligence on the U.S. workforce. By examining relationships between AIOE, median wages, education requirements, and projected employment growth, the project highlights how organizations can make more informed decisions about workforce planning, talent development, and technology investments. Findings suggest that higher AI exposure does not necessarily indicate declining employment demand, particularly among higher-paying and knowledge-based occupations. Businesses can use these insights to identify opportunities for employee upskilling, anticipate changes in occupational demand, and develop strategies that balance AI adoption with workforce sustainability. Ultimately, this project demonstrates how data analytics can help organizations evaluate emerging labor-market trends and support evidence-based planning.

Data Profiling and Observations
#1

File Name: occupation.xlsx

Description:U.S Bureau of Labor Statistics 2024

Dataset Details: 1119 Rows & 16 Columns

Size: 398 KB

Source: (U.S) Bureau of Labor Statistics - Occupational projections, 2024–2034, and worker characteristics, 2024 (Employment in thousands)

#2

AIOE Dataset

FIle Name: AIOE_DataAppendix.xlsx

Description: AIOE Data- Appendix A

Dataset details: 775 rows & 3 columns Size: 167 KB

Source: GitHub- AIOE Data

Data Table/ Schema

Name	Description
Employment title	The name of the occupation represented in the dataset. This field identifies the specific career or occupational group being analyzed.
2024 National Employment Matrix (SOC) code	The Standard Occupational Classification (SOC) code assigned to each occupation by the BLS. It provides a standardized identifier that can also be used to match occupations across datasets.

Occupation Type	Indicates how the occupation is categorized with BLS Projections table, such as a detailed line-item occupation or occupational group.
AIOE	Measures how strongly an occupation's tasks are exposed to advances in AI . Higher scores indicate greater AI exposure, lower scores indicate less AI exposure.
Employment, 2024	Estimated number of people in the occupation in 2024. BLS reports employment figures in thousands of jobs.
Employment, 2034	Projected number of people in the occupation in 2034.
Employment distribution, percent, 2024	% of total U.S employment represented by occupation in 2024. It provides context for the occupation's relative size in the labor market.
Employment distribution, percent, 2034	Projected % of total U.S employment represented by occupation in 2034. It shows whether the occupation's relative share of employment is expected to change.
Employment change, numeric, 2024–34	Projected numeric change in employment from 2024-2034, reported in thousands of jobs. "+" values indicate job gains.
Employment change, percent, 2024–34	Projected % increase or decrease in employment from 2024-2034. This variable is especially useful for comparing expected job growth across occupations of different sizes.
Percent self employed, 2024	% of workers within an occupation who were self-employed in 2024. This provides insight into how much an occupation relies on independent or self-directed employment.
Occupational openings, 2024–34 annual average	Average number of job openings expected each year from 2024-2034. Includes openings created by employment growth & need to replace workers who leave occupation.
Median annual wage, dollars, 2024	Median annual wage for workers in the occupation in 2024. This variable helps evaluate the economic value and earning potential of each occupation.
Typical education needed for entry	Typical level of education workers needed to

	enter the occupation such as a bachelor's, or if no formal educational credential. This helps assess the educational requirements associated with different career occupations.

Data Process and Wrangling

- Downloaded occupation.xslx dataset from USBL site to chromebook. Uploaded dataset to Excel online.
- The original dataset consisted of 13 sheets. Each sheet is a breakout subset of table 1.2
- Table 1.2 was used as master source/tab and accordingly was copied and pasted into a blank workbook for purposes of cleaning.
- Line 1 and line 3 in table 1.2 were summaries of table 1.2.'s contents and were deleted to reduce redundancy.
- Column C cells reading " summary" in table 1.2 were filtered out/ deleted to avoid redundancy. "Summary" lines repeat what is to be analyzed as line items.
- Columns D-L in 1.2 were changed from general to numeric categories for cleaning & wrangling efficiency.
- Rows 1116-1119 were deleted because they contain footnotes. Columns N-P were deleted in table 1.2 because they are non-numeric, not helpful in the context of analysis.
- Located AIOE dataset; Appendix A is to be merged with Bureau of Labor Statistics data.
- AIOE data came readily available. Appeared correctly formatted. Pasted dataset into a new sheet in the same workbook as Bureau Labor Statistics table 1.2 for merging.
- Used Index-Match function to merge AIEO scores into new column O of table 1.2.

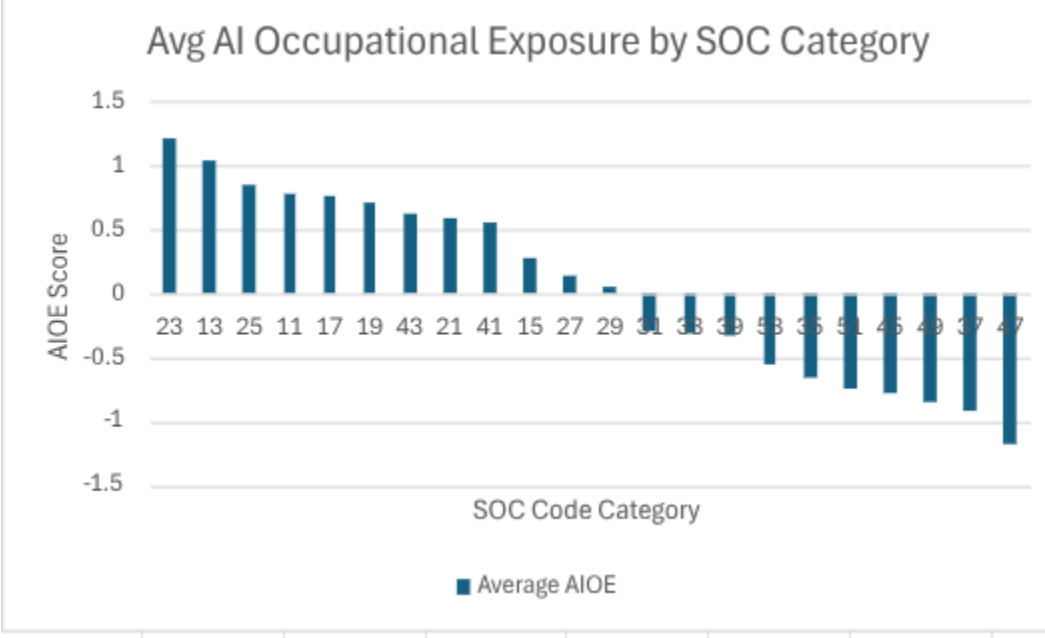
Data Cleaning

- Index - match processed - appendix A AIOE scores into table 1.2 as new column D.
- Some detailed occupations didn't have an original AIOE score, they had blanks.
- Instead of leaving those cells blank, they were estimated using the average score of similar occupations. First attempt: 4-digit SOC family
- If that family has no score, the broader 2-digit SOC group was used.

  Final data table contained : 14 columns, 832 rows in final data set

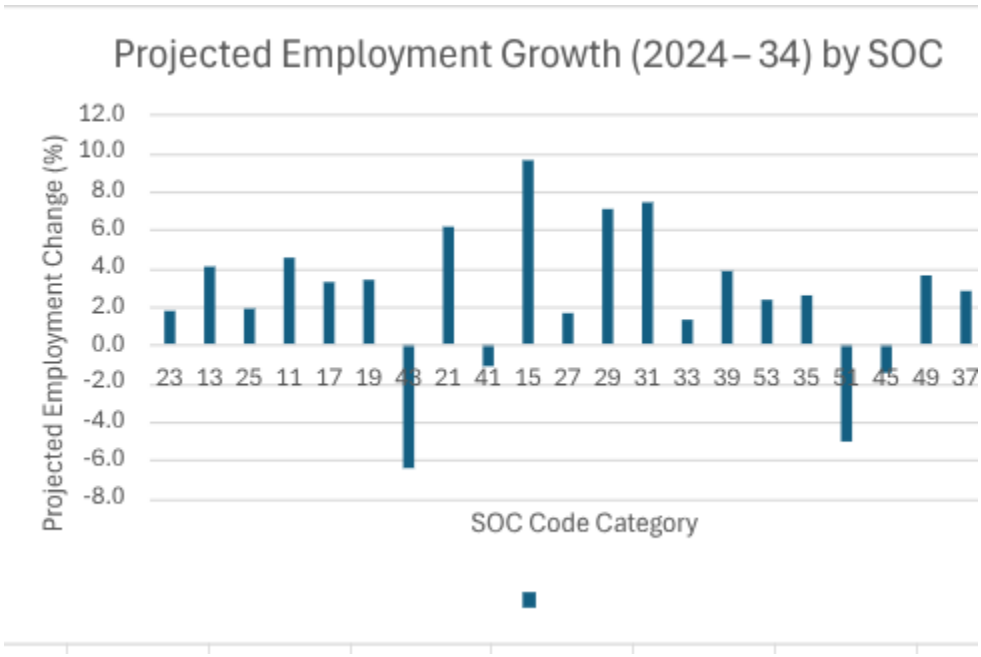
SOC Category Reference Key	
High AIOE/Cognitive	**Low AIOE/ Physical & Care Based**
23:Legal	29:Healthcare Prac
13:Business & Financ	31: Healthcare Support
15: Computer & Math	35: Food Prep & Service
25: Educational/Inst.	47: Construction
43: Office & Admin	51: Production

Exploratory Data Analysis



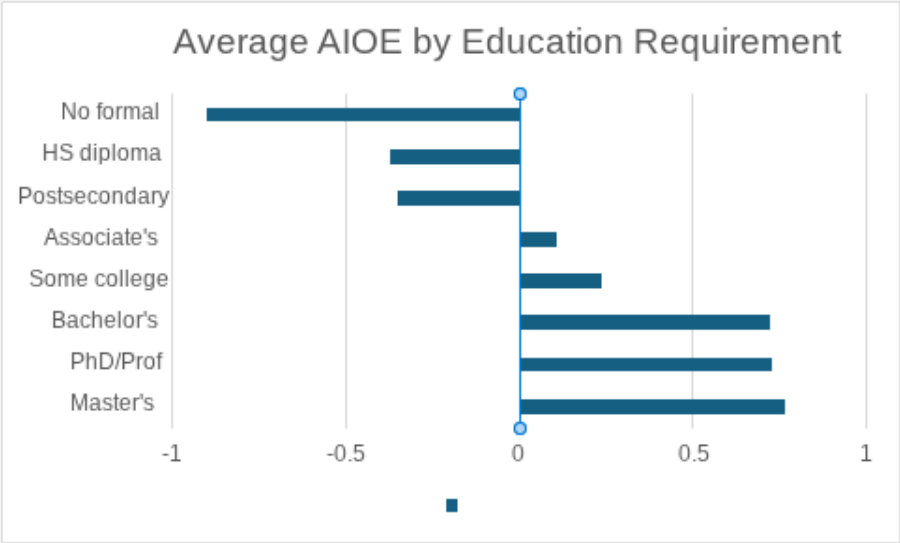
The first visualization depicted AIOE by SOC and examined which occupational groups have lower or higher AIOE exposure. Average AIOE scores were found to vary substantially across SOC major occupational groups. For example Legal occupations (SOC 23) and Business and Financial Operations occupations (SOC 13) had the highest average AIOE scores, while several occupational groups characterized by more physical or manual work appear toward the lower end of the distribution. Overall, the visualization suggested greater AI exposure among occupations involving information processing, language, analysis, and other

cognitive tasks, while occupations requiring more physical, hands-on,or environment-dependent activities tend to demonstrate lower exposure.



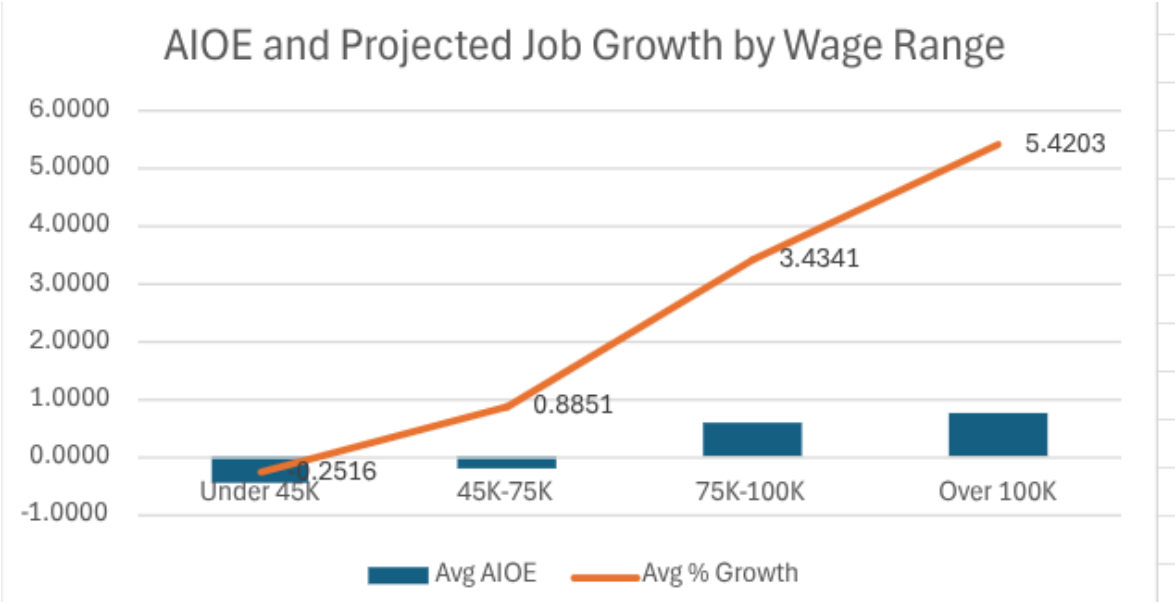
The second visualization depicted how projected employment growth varies considerably across SOC occupational groups. Computer and Mathematical Occupations (SOC 15- such as data scientists, operations research analysts, and computer systems analysts) show the highest average projected growth among the categories shown at approximately 9.6%, showing the computer and mathematical category is projected to grow strongly even though AI is increasingly relevant to many of the tasks performed in those occupations. Legal

Occupations (SOC 23) appeared to have the highest average AIOE score overall (~1.21) .



Visualization 3 illustrates how the average AIOE for various occupations shows a noticeable relationship with the typical education required for entry. Occupations requiring a master's degree have the highest average AIOE score (~0.77), followed closely by doctoral/professional degree (~0.73) and bachelor's degree occupations (~0.72). In comparison, occupations requiring some college but no degree (~0.24) or an associate's degree (~0.11) show substantially lower positive exposure. Overall, the pattern suggests that occupations requiring higher levels of formal education tend to possess greater overlap with current AI capabilities.

These results challenge the assumption that workers in occupations requiring less education necessarily face the greatest AI exposure. In this dataset, the strongest AI overlap appears among occupations requiring bachelor's and advanced degrees. However, a higher AIOE score represents greater exposure or task overlap with AI capabilities, not a direct probability of job displacement. Therefore, projected employment growth needs to be examined before determining occupational groups' employment risk.



Visualization 4 portrays how projected job growth by wage category relates to AIOE. There appeared to be two simultaneous upward trends as wage levels increased. Average AIOE increased substantially, moving from negative values in the two lower-wage groups to positive values in the two higher-wage groups. At the same time, projected employment growth rises from slightly negative for occupations under $45K to 5.42% for occupations paying over $100K. These trends complicate the idea that greater AI exposure automatically means greater job displacement. Based on the data sets analyzed, the higher-wage groups have both greater average AI exposure and stronger projected employment growth.

Challenges and Solutions

Two key challenges arose during the data preparation and visualization stages of this project. First, the Tableau dashboard initially appeared overcrowded, making it difficult to interpret the visualizations. Previewing the published dashboard helped identify layout issues and guided adjustments to container spacing and chart organization resulted in a cleaner, more accessible presentation. Additionally, missing AIOE scores presented challenges during data preparation. To address these gaps, missing values were imputed using average AIOE scores

from the corresponding 4-digit SOC occupational groups or broader 2-digit SOC categories, allowing for a more complete dataset while maintaining consistency across occupational classifications.

Conclusion

The analysis indicates that AI exposure varies substantially across occupations and is especially pronounced among occupations requiring higher levels of education. Average AIOE was highest among occupations requiring a Master's degree (0.769), followed by PhD/professional degrees (0.732) and Bachelor's degrees (0.724). In comparison, occupations requiring no formal education had an average AIOE of −0.895, suggesting lower exposure to AI.

However, high AI exposure should not automatically be interpreted as high risk of job elimination. The employment-growth analysis illustrates this distinction. Computer and Mathematical occupations (SOC 15), for example, have projected employment growth of approximately 9.6% until 2034, the highest among the SOC categories analyzed, despite knowledge-based professional work of this occupational group being relatively exposed to AI. Therefore, AIOE appears to measure the potential for AI to affect occupational tasks more effectively than it predicts whether an occupation will disappear.

The analysis suggests that the relationship between earnings, job growth, and AI exposure is complex rather than directly inverse. Overall, AIOE and projected employment growth had a weak negative correlation of approximately −0.25, meaning greater AI exposure was only weakly associated with lower projected growth across individual occupations.

The wage analysis revealed an additional pattern. Occupations earning under $45,000 had an average AIOE of −0.46 and average projected growth of approximately −0.25%, while

occupations earning over $100,000 had the highest average AIOE (0.75) but also the highest average projected growth (5.42%).

Specific occupational categories demonstrate why this distinction matters. Computer and Mathematical occupations (SOC 15) are projected to grow 9.6%, while Healthcare Practitioners and Technical occupations (SOC 29) and Healthcare Support occupations (SOC 31) are projected to grow 7.1% and 7.4%, respectively. In contrast, Office and Administrative Support occupations (SOC 43) are projected to decline 6.4%, while Production occupations (SOC 51) are projected to decline approximately 5.0%. These results suggest that higher earnings and greater AI exposure can coexist with strong employment growth. AI exposure alone therefore does not determine an occupation's economic outlook.

The findings suggest that career resilience should be evaluated using multiple indicators together: AIOE, projected employment growth, wages, education requirements, and occupational category. For example, Computer and Mathematical occupations (SOC 15) show very strong projected growth (9.6%), while the education analysis shows that Bachelor's- and graduate-degree occupations generally have relatively high AI exposure. This represents a career area that may be highly exposed to AI-driven changes without necessarily being economically vulnerable. By comparison, Office and Administrative Support (SOC 43) has projected employment decline of −6.4%, while Production occupations (SOC 51) are projected to decline approximately −5.0%. Occupations that combine declining employment projections with substantial automation exposure would warrant greater attention than occupations that simply have a higher AIOE. The data ultimately suggests that individuals can and should distinguish between exposure and vulnerability. High AIOE may indicate that the *tasks within a job* are likely to change, while the combination of high exposure, weak occupational growth projection , and weaker economic outcomes provides a stronger indication of potential career vulnerability.

Overall, this exploratory analysis suggests that AI exposure does not automatically translate into job displacement. AIOE was considerably higher among occupations requiring Bachelor's and graduate degrees, demonstrating that AI exposure extends strongly into highly educated, knowledge-based work. At the same time, several occupational categories showed strong employment projections, including Computer and Mathematical (9.6%), Healthcare Support (7.4%), Healthcare Practitioners and Technical (7.1%), and Community and Social Service occupations (6.2%). Conversely, Office and Administrative Support (−6.4%) and Production occupations (approximately −5.0%) showed substantial projected declines. The wage analysis further demonstrated that occupations earning over $100,000 had both the highest average AIOE (0.75) and the strongest average projected growth (5.42%).

The findings of this analysis suggest that automation vulnerability is multidimensional. Evaluating AI exposure alongside wages, education requirements, occupational category, and projected employment growth provides a more meaningful assessment of career resilience.